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COMMON QUARTERLY EXAMINATION - 2025-26

MATHEMATICS

Time Allowed : 3.00 Hours] [Max. Marks : 90

PART - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives 20 x 1 = 20
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If $|adj(adj\ A)| = |A|^9$, then the order of the square matrix A is

(1) 2 (2) 3 (3) 4 (4) 5

2. If adj $A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$, adj $B = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$, then adj (AB) is

(1) $\begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix}$ (2) $\begin{bmatrix} -7 & -1 \\ 7 & -9 \end{bmatrix}$ (3) $\begin{bmatrix} -6 & -2 \\ 5 & -10 \end{bmatrix}$ (4) $\begin{bmatrix} -7 & 7 \\ -1 & -9 \end{bmatrix}$

3. If z is a non zero complex number, such that $2iz^2 = \bar{z}$ then $|z|$ is

(1) 3 (2) 2 (3) 1 (4) 3 ½

4. If z is a complex number such that $z \in C \setminus R$ and $z + \frac{1}{z} \in R$, then $|z|$ is

(1) 2 (2) 1 (3) 3 (4) 0

5. If α, β, and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is

(1) $-\frac{p}{r}$ (2) $-\frac{q}{r}$ (3) $-\frac{q}{p}$ (4) $\frac{q}{r}$

6. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1 = 0$ is

(1) ∞ (2) 1 (3) 2 (4) 4

7. The value of $\sin^{-1}(\cos x)$, $0 \le x \le \pi$ is

(1) $x - \frac{\pi}{2}$ (2) $\pi - x$ (3) $x - \pi$ (4) $\frac{\pi}{2} - x$

8. $\sin^{-1}\frac{3}{5} - \cos^{-1}\frac{12}{13} + \sec^{-1}\frac{5}{3} - \text{cosec}^{-1}\frac{13}{12}$ is equal to

(1) $\tan^{-1}\frac{12}{65}$ (2) 0 (3) π (4) 2π

9. If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to

(1) $[-2, -\sqrt{2}]$ (2) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$ (3) $[-1, 1]$ (4) $[\sqrt{2}, 2]$

10. The equation $\tan^{-1} x - \cot^{-1} x = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ has

(1) two solutions (2) no solution (3) unique solution (4) infinite number of solutions

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11. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

(1) $\frac{4}{\sqrt{3}}$ (2) $\frac{4}{3}$ (3) $\frac{3}{2}$ (4) $2\sqrt{3}$

12. If P(x, y) be any point on $16x^2 + 25y^2 = 400$ with foci $F_1(3, 0)$ and $F_2(-3, 0)$ then $PF_1 + PF_2$ is

(1) 6 (2) 10 (3) 12 (4) 8

13. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ parallel to the straight line $2x - y = 1$. One of the points of contact of tangents on the hyperbola is

(1) $\left(\frac{-9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (2) $\left(\frac{9}{2\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ (3) $(3\sqrt{3}, -2\sqrt{2})$ (4) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

14. If $[\vec{a}, \vec{b}, \vec{c}] = 1$, then the value of $\frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{(\vec{c}\times\vec{a})\cdot\vec{b}} + \frac{\vec{b}\cdot(\vec{c}\times\vec{a})}{(\vec{a}\times\vec{b})\cdot\vec{c}} + \frac{\vec{c}\cdot(\vec{a}\times\vec{b})}{(\vec{c}\times\vec{b})\cdot\vec{a}}$ is

(1) 2 (2) 3 (3) 1 (4) -1

15. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a}\times(\vec{b}\times\vec{c}) = \frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(1) π (2) $\frac{\pi}{4}$ (3) $\frac{3\pi}{4}$ (4) $\frac{\pi}{2}$

16. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then

(1) $c > 0$ (2) $0 < c < 1$ (3) $c = \pm 3$ (4) $c = \pm\sqrt{3}$

17. $\text{cis}\frac{28\pi}{5} =$

(1) $\text{cis}\frac{-2\pi}{5}$ (2) $\text{cis}\frac{2\pi}{5}$ (3) $\text{cis}\frac{3\pi}{5}$ (4) $\text{cis}\frac{-3\pi}{5}$

18. Domain of $\text{cosec}^{-1} x$

(1) R\(-1,1) (2) R\{-1,1} (3) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (4) R-{0}

19. $y = mx + c$ is a tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then

(1) $c = \frac{a}{m}$ (2) $c^2 = a^2m^2 + b^2$ (3) $c = am + b$ (4) $c^2 = a^2m^2 - b^2$

20. $\vec{a}.(\vec{b}\times\vec{c}) =$

(1) $\vec{a}.(\vec{c}\times\vec{b})$ (2) $\vec{c}.(\vec{b}\times\vec{a})$ (3) $-\vec{b}.(\vec{c}\times\vec{a})$ (4) $(\vec{c}\times\vec{a}).\vec{b}$

PART - II

7 x 2 = 14

1. Answer any 7 questions
2. Each question carries 2 marks
3. Question number 30 is compulsory

21. Find the product $\frac{3}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\cdot 6\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$ in rectangular from.

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22. If $x^2 + 2(k+2)x + 9k = 0$ has equal roots, find k.

23. Is $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$ true? Justify your answer.

24. Find the value of $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)$

25. Obtain the equation of the circle for which (3,4) and (2, – 7) are the ends of a diameter.

26. Find the equations of tangent and normal to the parabola $x^2 + 6x + 4y + 5 = 0$ at (1, – 3).

27. Construct a cubic equation with roots $2, \frac{1}{2}$ and 1.

28. If $\vec{a} = -3\hat{i} - \hat{j} + 5\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{c} = 4\hat{j} - 5\hat{k}$, find $\vec{a} \cdot (\vec{b} \times \vec{c})$.

29. Find the length of the perpendicular from the point (1, – 2, 3) to the plane $x - y + z = 5$.

30. If A is symmetric, prove that adj A is also symmetric.

PART - III

1. Answer any 7 questions
2. Each question carries 3 marks
3. Question number 40 is compulsory

7 x 3 = 21

31. Find the rank of the matrix by row reduction method: $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

32. If $|z| = 3$, show that $7 \le |z + 6 - 8i| \le 13$.

33. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solutions.

34. Find the domain of $\sin^{-1}(2 - 3x^2)$

35. If $\cot^{-1}\left(\frac{1}{7}\right) = \theta$, find the value of $\cos\theta$.

36. Find the equation of the ellipse in case given below: length of latus rectum 8, eccentricity $= \frac{3}{5}$ and major axis on x -axis.

37. Find the equations of tangent and normal to the parabola $x^2 + 6x + 4y + 5 = 0$ at (1, – 3).

38. With usual notations, in any triangle ABC, prove by vector method that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.

39. The vertices of ΔABC are A(7,2,1), B(6,0,3), and C(4,2,4). Find $\angle ABC$.

40. Find the value of $\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

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PART - IV

1. Answer all the questions
2. Each question carries 5 marks

7 x 5 = 35

41. a) An amount of ₹65,000 is invested in three bonds at the rates of 6%, 8% and 10% per annum respectively. The total annual income is ₹4,800. The income from the third bond is ₹600 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method.)

(OR)

b) If $A=\begin{bmatrix}-4 & 4 & 4\\ -7 & 1 & 3\\ 5 & -3 & -1\end{bmatrix}$ and $B=\begin{bmatrix}1 & -1 & 1\\ 1 & -2 & -2\\ 2 & 1 & 3\end{bmatrix}$, find the products AB and BA and hence solve the system of equations $x-y+z=4, x-2y-2z=9, 2x+y+3z=1$.

42. a) Given the complex number $z=3+2i$, represent the complex numbers z, iz, and z+iz in one Argand diagram. Show that these complex numbers form the vertices of an isosceles right triangle.

(OR)

b) If $\omega \neq 1$ is a cube root of unity, show that the roots of the equation $(z-1)^3+8=0$ are $-1, 1-2\omega, 1-2\omega^2$

43. a) If $2+i$ and $3-\sqrt{2}$ are the roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$. Find all roots.

(OR)

b) Solve : $6x^4-35x^3+62x^2-35x+6=0$

44. a) Evaluate $\sin\left[\sin^{-1}\left(\frac{3}{5}\right)+\sec^{-1}\left(\frac{5}{4}\right)\right]$

(OR)

b) Prove that $\tan^{-1}x+\tan^{-1}\frac{2x}{1-x^2}=\tan^{-1}\frac{3x-x^3}{1-3x^2}, |x|<\frac{1}{\sqrt{3}}$.

45. a) Find the equation of the circle passing through the points (1,1), (2, – 1) and (3,2).

(OR)

b) Find the vertex, focus, equation of directrix and length of the latus rectum: $x^2-2x+8y+17=0$

46. a) A tunnel through a mountain for a four lane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16m, and the height at the edge of the road must be sufficient for a truck 4m high to clear if the highest point of the opening is to be 5m approximately. How wide must the opening be?

(OR)

b) Prove by vector method that $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$

47. a) Show that the lines $\frac{x-3}{3}=\frac{y-3}{-1}, z-1=0$ and $\frac{x-6}{2}=\frac{z-1}{3}, y-2=0$ intersect. Also find the point of intersection.

(OR)

b) Find the vector and cartesian equation of plane containing $\frac{x-2}{2}=\frac{y-2}{3}=\frac{z-1}{3}$ and parallel to the line

$\vec{r}=(\hat{i}+\hat{j}+2\hat{k})+s(3\hat{i}+2\hat{j}+\hat{k})$

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